✅ Task 1: Onboarding Strategy for Port of Los Angeles (POLA)

Objective

Enable the Port's security operations to transition from primarily manual and reactive surveillance to a **drone-assisted, proactive response system**, improving incident response time, coverage, and operational reliability.

Phase 1: Current State Discovery & Site Readiness (Weeks 1–2)

Goal:

Understand existing workflows, constraints, and success criteria before deployment.

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1. Current Workflow & Expectation Discovery

- Understand how security is currently managed:
  - Patrol methods (guards, vehicles, CCTV)
  - Incident detection and escalation process
  - Average response time
- Identify operational gaps:
  - Delays in response
  - False alarms
  - Visibility limitations
- Align on success criteria with stakeholders:
  - Faster response time
  - Reduced manual patrol effort
  - Improved coverage
  - Reliable day-to-day operation

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2. Blind Spot & Low-Light Mapping

- Identify areas with:
  - Limited CCTV visibility (e.g., behind container stacks)
  - Poor lighting / night-time visibility challenges
  - High intrusion risk zones

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3. Site Survey & Dock Placement

- Identify optimal locations for drone docks (e.g., DJI Dock 2):
  - Clear sky view for GPS
  - Stable power supply (220V)
  - Strong 4G/5G connectivity
  - Minimal obstruction from cranes, containers, and vessels

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4. Safety & Regulatory Planning

- Define no-fly zones (cranes, vessels, restricted areas)
- Establish emergency landing protocols
- Ensure compliance with aviation regulations (FAA) and port safety guidelines

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Why Drone-Based Security is Needed (Context Alignment)

While the Port already uses CCTV and sensor systems, these are **primarily passive monitoring tools** with limitations such as:

- Fixed camera angles leading to blind spots
- Reduced effectiveness in low-light or obstructed environments
- Dependence on human monitoring, causing delayed response

FlytBase enables:

- **On-demand aerial visibility** to eliminate blind spots
- **Automated response to triggers** without manual delay
- **Real-time tracking and situational awareness**

👉 This shifts operations from **“monitoring incidents” to “actively responding to them”**

💥 **Key Principle:**

> The focus is not just deployment, but ensuring predictable performance under real-world conditions, including environmental and operational edge cases.

Phase 2: Integration & Automation (Weeks 3–4)

Goal:

Integrate FlytBase into existing security infrastructure with minimal disruption.

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1. Unified Monitoring ("One-Screen Setup")

- Integrate drone video feed into existing VMS
- Drone appears as an additional camera feed
  👉 No need for operators to switch systems

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2. Automated Response Workflows

Example:

- Fence breach detected at a specific zone
  → Drone auto-launches within defined SLA
  → Live video streamed to SOC

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3. Event-Based Trigger Integration

- Integrate with:
  - Fence sensors
  - Motion detection systems
  - Optional: AIS (vessel tracking)

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4. System Validation

- Validate:
  - Video latency
  - Command responsiveness
  - Network reliability across coverage zones

Phase 3: Pilot Deployment & Reliability Testing (Weeks 5–6)

Goal:

Test system performance under real operational conditions.

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1. Controlled Pilot Deployment

- Start with a high-risk zone
- Execute:
  - Scheduled patrol missions
  - Simulated incident scenarios

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2. Edge Case & Failure Testing

- Signal loss scenarios
- High wind conditions
- Low battery mid-mission
- False alarm triggers

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3. Safety Logic Validation

- Auto return-to-home
- Mission abort conditions

- Manual override capability

💥 **Key Focus:**

> Identify and resolve failure scenarios early to ensure reliability in real-world operations.

Phase 4: Training & Operational Enablement

Goal:

Enable the security team to operate the system independently.

Role-Based Training

Security Personnel

- Mission execution
- Live monitoring
- Camera control (gimbal tracking)

Supervisors

- Oversight and escalation handling

IT/Admin

- System monitoring and basic troubleshooting

Training Approach

- Hands-on sessions
- Scenario-based drills
- Real-world simulations

Phase 5: Go-Live & Scale (Week 7 onwards)

Goal:

Expand from pilot to full deployment.

Activities

- Extend drone coverage across the port
- Enable:
    - Scheduled patrols
    - Event-triggered missions

Success Metrics

- Response time improvement
- Mission success rate
- Reduction in manual patrol effort
- User adoption

Phase 6: Ongoing Monitoring & Optimization

Goal:

Ensure long-term adoption and continuous improvement.

Approach

- Regular performance reviews
- Optimize patrol routes and trigger logic
- Reduce false positives
- Incorporate operator feedback

✅ Task 2: Customer Health Score Framework (Final Version)

Objective

Provide a structured way to evaluate customer satisfaction, system performance, and long-term retention risk.

Customer Health Model (Balanced Score Approach)

Customer health is evaluated across **four equally critical dimensions**, each scored independently out of 100.

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1. Mission Success Rate (Reliability)

- % of missions completed successfully
- Indicates system stability and trust

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2. Response SLA (Performance)

- Time from alert trigger → live video feed
- Measures operational effectiveness

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3. Team Adoption (Usage)

- % of security personnel actively using the system
- Reflects real-world value and usability

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4. Safety Compliance (Risk Control)

- Adherence to no-fly zones and safety rules
- Ensures regulatory and operational safety

Measurement Approach

- Mission Success → platform logs (success vs failure)
- Response SLA → event timestamps (trigger → video feed)
- Adoption → user activity logs (logins, interactions)
- Safety → flight logs and geofence violations

How This Will Be Used

- Each metric is monitored independently
- Any metric falling below a threshold (e.g., <80) triggers investigation

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💥 **Key Insight:**

> A balanced model avoids masking critical issues, as failure in any one dimension can significantly impact customer trust and long-term retention.

Operational Interpretation Examples

- High adoption + low reliability → trust issue
- High reliability + low adoption → training/usability gap
- Strong performance but safety violations → compliance risk

Task 3: Product Adoption Strategy & Email Sequence

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0. Assumptions & Validation Approach

> The following strategy is based on typical port security environments and will be validated during discovery.

Assumptions:

- Existing surveillance includes CCTV and manual patrols
- Drone workflows are either new or underutilized
- Potential prior exposure to drone systems may influence user confidence

Validation:

- Stakeholder discussions (CSO, Ops, IT)
- Workflow observation
- Usage and incident data analysis

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1. Problem Diagnosis: The Trust & Friction Barrier

Low adoption is not a knowledge issue—it is a **behavioral and operational gap**.

🔹 A. Legacy Workflow Inertia

- Security teams rely on:
  - CCTV monitoring
  - Manual patrol verification
- These are deeply embedded habits

> Introducing drones adds perceived cognitive load unless seamlessly integrated.

🔹 B. Trust Deficit (Possible Prior Experience)

- If previous drone systems:
  - Required manual piloting
  - Were unreliable
  - Were hard to operate

→ Users may have **low confidence in automation**

> Adoption depends on rebuilding trust through consistent, predictable behavior.

🔹 C. Workflow Friction

- Switching between systems (CCTV → Drone interface)
- Multi-step execution
- Unclear operational benefit

> Even small friction points prevent adoption in high-pressure environments.

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2. Strategic Positioning: FlytBase as a "Failsafe Verification Layer"

FlytBase should be positioned as:

> A **reliable, low-friction verification layer** that integrates into existing security workflows.

🔹 If CCTV Exists:

- FlytBase provides:
  - Rapid visual confirmation
  - Improved situational awareness

🔹 If Prior Drone Systems Exist:

- Position FlytBase as:
  - More workflow-integrated

    - Easier to operate
    - Focused on operational reliability

🔹 **Core Value:**

- Reduce ambiguity
- Reduce manual verification effort
- Improve response consistency

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3. Re-Engagement Strategy

Step 1: Identify Real Friction (Data + Observation)

- Analyze:
    - Mission usage patterns
    - Drop-off points
- Conduct:
    - Stakeholder interviews
    - **Workflow shadowing (critical step)**

Observe real behavior instead of relying only on reported feedback.

Step 2: Define a Simple, High-Impact Workflow

👉 Focus:
Alert Verification + Perimeter Patrol

- Pre-configured missions
- Minimal user interaction

Step 3: Pilot-Based Validation

- Select:
    - 1–2 high-friction zones
- Compare:

    - Current method vs drone-assisted verification

Step 4: Demonstrate Measurable Value

Track:

- Time to verify alerts
- Reduction in manual patrols
- Coverage improvement
- Time to initiate mission

🔥 Added Metric (Strong Differentiator)

Time to First Launch

> Time taken by operator to initiate drone mission after alert.

- High → friction exists
- Low → workflow is effective

Step 5: Reduce Operational Friction

- Pre-configured missions
- Simplified UI flows
- Minimal steps

Step 6: Build Trust Through Reliability

- Ensure:
    - Consistent mission execution
    - Predictable system behavior
- Design workflows that:
    - Do not require manual piloting
    - Handle failures gracefully

Step 7: Align with Leadership

Translate into:

- Faster response
- Reduced patrol effort
- Improved operational coverage

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4. Email Outreach Sequence

Email 1 — Friction Discovery (Ops Manager)

Subject: Aligning FlytBase with current response workflows

Hi [Name],

I've been reviewing usage patterns and noticed the team is still relying heavily on manual patrols for certain verification tasks.

In similar environments, this usually points to workflow friction rather than a technology limitation.

From your perspective, are there specific challenges—such as transition steps from CCTV to drone workflows or usability concerns—that are affecting adoption?

Happy to align on simplifying this for the team.

Best,
Kaushal Raj

Email 2 — End User Enablement

Subject: Simplifying alert verification during patrols

Hi Team,

In many cases, verifying unclear alerts requires manual patrols, which can slow down response time.

We've configured a simplified workflow that allows quick drone-based verification with minimal interaction.

This can help reduce unnecessary patrol effort and improve response speed.

Happy to walk through this with the team.

Best,
Kaushal Raj

Email 3 — Leadership Alignment

Subject: Improving response efficiency and operational coverage

Hi [Name],

We’ve identified an opportunity to improve how alerts are verified, particularly in areas where visibility is limited.

Currently, this often requires manual intervention, which increases response time and operational effort.

By introducing a simplified drone-based verification workflow, we can improve response speed and consistency.

Would be useful to align on key success metrics and priority areas.

Best,
Kaushal Raj

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5. Action Plan for Persistent Low Adoption

🔹 1. Workflow Shadowing

- Observe real-time operations
- Identify:
  - Where users hesitate
  - Where process breaks

🔹 2. Workflow Simplification

- Reduce steps
- Adapt UI to user environment
- Align with existing tools

🔹 3. Reliability Reinforcement

- Ensure consistent behavior
- Minimize need for manual intervention
- Provide predictable outcomes

🔹 4. Champion-Based Adoption

- Identify active users
- Enable peer learning

> “Adoption is driven by trust, workflow fit, and reliability. By reducing operational friction, validating value through real-world usage, and embedding FlytBase into existing workflows, we can transition it from a new tool to a standard part of security operations.”

Task 4: Onboarding Presentation

YouTube Link: <https://youtu.be/VylLmcZRYNw>

Task 5: Technical Integration Plan (VMS & SOC)

Objective

Integrate FlytBase into the Port of Los Angeles (POLA) security setup so that drones become a **natural part of existing monitoring and response workflows**, without adding complexity for the security team.

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1. Integration Logic: Trigger → Action → Visibility

Instead of manual drone operation, the system should work as an **automatic response layer**.

Flow:

Sensor / CCTV detects issue

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Existing system (VMS / alarm)

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Trigger sent to FlytBase

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Drone launches automatically

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Live video appears in same monitoring screen

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Operator takes decision

👉 Key idea:
No manual flying, no switching between systems.

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2. Key Integration Components

2.1 Trigger Integration

- Existing systems (CCTV, fence sensors) send a signal to FlytBase
- This can be done using simple API/webhook calls (similar to workflow tools)

👉 Example:

- Fence breach detected → trigger sent → drone launches automatically

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2.2 Video Integration

- Drone video is shown inside the existing monitoring system (VMS)
- It appears like another camera feed (e.g., “Drone Camera”)

👉 Outcome:

- Security team does not need to learn new software
- They use the same screen they already trust

2.3 Command & Control

- System runs automatically
- But operators can:
    - Start a mission manually
    - Stop or override if needed

👉 Important:
Always keep human control available

2.4 Data & Incident Logging

- After each mission:
    - Video and key details are saved
    - Can be linked to incident reports

👉 Helps in:

- Investigation
- Compliance
- Record keeping

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3. Reliability & Fail-Safe (Most Important)

From real-world experience, systems must **fail safely**.

Key safeguards:

- **Connection Loss**
  → Drone returns to base automatically
- **System Failure**
  → Alert shown to operators
  → They switch to manual patrol
- **Manual Override**
  → Operator can stop mission anytime

👉 Key principle:
System should never fail silently

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4. Key Challenges & Solutions

1. Network Issues

Ports have signal challenges (containers, structures)

→ Solution:

- Test connectivity in advance
- Use backup network options

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2. Too Many Alerts

Too many triggers reduce trust

→ Solution:

- Filter alerts

- Use only important triggers

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3. Low Adoption

If system feels complicated, people won’t use it

→ Solution:

- Keep everything in one screen
- Reduce steps
- Use pre-configured missions

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4. Safety

Risk of flying in restricted areas

→ Solution:

- Define no-fly zones
- Set emergency rules

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5. Implementation Approach

- Start with small pilot area
- Test trigger → launch → video flow
- Fix issues
- Expand gradually

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6. Success Metrics

- Faster response time
- Reduction in manual patrols
- Successful drone missions

- Team actively using the system

“The goal is to make drones a simple and reliable extension of existing security operations, not an additional system that increases complexity.”

Task 6: Customer Growth Plan

1. Vision: From Security Tool to Operational Visibility Layer

The goal is to evolve FlytBase from a security-focused deployment into a broader operational visibility layer across the port.

Instead of being used only for surveillance, drones can support:

- **Inspections**
- **Monitoring**
- **Operational decision-making**

👉 **This increases both platform value and long-term adoption.**

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2. Expansion Use Cases

2.1 Infrastructure Inspection

Use Case:

- **Inspection of cranes, lighting towers, container stacks**

How it works:

- **Scheduled autonomous missions capture high-resolution and thermal data (via compatible sensors)**

Value:

- **Reduces manual inspection effort**
- **Improves safety (less work at height)**
- **Enables early issue detection**

2.2 Environmental & Hazard Monitoring

Use Case:

- **Oil spills, heat hotspots, gas leak indicators**

How it works:

- **AI-assisted detection using thermal imaging and visual analysis**

Value:

- **Faster detection of risks**
- **Supports compliance and safety standards**

2.3 Logistics & Traffic Monitoring

Use Case:

- **Monitor truck queues and yard congestion**

How it works:

- **Live drone feeds integrated with existing monitoring systems**

- **Optional analytics (e.g., vehicle movement tracking / identification)**

Value:

- **Improves traffic flow decisions**
- **Reduces congestion and delays**

2.4 Emergency Response Support

Use Case:

- **Fire, intrusion, hazardous events**

How it works:

- **Automated or on-demand drone deployment**
- **Real-time aerial visibility**

Value:

- **Faster response**
- **Better situational awareness**

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3. Implementation Roadmap

Phase 1 (0–3 Months): Stabilize & Optimize

- **Ensure current security workflows are reliable**
- **Improve trigger → response → video flow**
- **Build operator confidence**

Phase 2 (3–6 Months): Pilot Expansion

- **Select one use case (e.g., crane inspection)**
- **Run pilot in one zone**
- **Compare with existing manual process**

Phase 3 (6–12 Months): Scale & Integrate

- **Expand to multiple zones**
- **Standardize workflows**
- **Integrate outputs into operational processes**

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4. Adoption Strategy (Critical for Growth)

- **Keep workflows simple**
- **Reuse existing monitoring interfaces**
- **Use pre-configured missions**
- **Demonstrate clear operational value**

👉 **Expansion should feel like an extension, not a new system**

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5. Success Metrics

- **Increase in total drone mission volume**
- **Reduction in inspection time (target: ~40–60%)**
- **Reduction in manual patrol/inspection effort**
- **Faster incident response time**
- **Cross-team adoption (security → operations → maintenance)**

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6. Growth Execution Strategy

- **Conduct periodic stakeholder reviews**
- **Identify high-impact workflows**
- **Demonstrate measurable improvements**
- **Enable internal champions across teams**

“The goal is not to introduce new complexity, but to extend an already trusted system into more workflows—so drones become a natural part of daily operations, not a separate tool.”